Changin Oh

PhD Candidate | Mathematical Oncology, Computational Biology, Scientific Machine Learning

https://changinoh.github.io c4oh@torontomu.ca

Education

Toronto Metropolitan University Toronto, ON

PhD in Mathematics 2023 – Present

- Thesis: “Reliable Mathematical Modelling in Oncology: Model Formulation, Parameter Estimation, and Data-Driven Discovery”
- Supervisor: Kathleen Wilkie

York University North York, ON

MA in Mathematics 2021 – 2023

- Thesis: “Linear Spectral Unmixing Algorithms for Abundance Fraction Estimation in Spectroscopy”
- Supervisor: Iain Moyles

University of Western Ontario London, ON

BSc in Mathematics 2009 – 2017

- Minor in Japanese Studies
- Dean’s Honour List for 2009 – 2011

Publications

Peer-Reviewed Articles

1. M. Oh, **C. Oh**, T. Ghosh, and E. Tabassum, “Selective effects of backbone cyclization and disulfide bonding as global covalent constraints on the conformational ensemble of sunflower trypsin inhibitor-1,” *The Journal of Physical Chemistry B*, Jun 2026.
2. **C. Oh** and K. P. Wilkie, “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection,” *Mathematical Biosciences*, vol. 396, p. 109685, Jun 2026.
   - *Featured in Mathematical Oncology Newsletter (research highlight)*
3. M. Oh, **C. Oh**, and E. Tabassum, “Covalent: Interpretable and discriminative collective variables reveal ligand-dependent switching in human cellular retinol-binding protein 2,” *Journal of Chemical Theory and Computation*, vol. 21, no. 23, pp. 12356–12370, Dec 2025.

4. M. Oh, **C. Oh**, and D. F. Weaver, "Effect of cholesterol on the structure of networked water at the surface of a model lipid membrane," *The Journal of Physical Chemistry B*, vol. 124, no. 18, pp. 3686–3694, May 2020.
5. M. Oh, **C. Oh**, M. Gupta, and D. F. Weaver, "Decoding interfacial water orientation to predict surface charge density on a model sheet using a deep learning algorithm," *The Journal of Physical Chemistry C*, vol. 124, no. 4, pp. 2574–2582, Jan 2020.
6. M. Oh, M. Gupta, **C. Oh**, and D. F. Weaver, "Understanding the effect of nanoconfinement on the structure of water hydrogen bond networks," *Physical Chemistry Chemical Physics*, vol. 21, no. 47, pp. 26237–26250, Dec 2019.

Blog Posts

1. A. Köhn-Luque, J. Nardini, **C. Oh**, L. Podina, G. Powathil, E. Rutter, J. West, and K. Wilkie, "Learning dynamics with neural networks," *Mathematical Oncology Blog*, Mar 2025.

Awards and Prizes

Ontario Graduate Scholarship .... 2026 – 2027
Awarded as a prestigious, competitive provincial scholarship to graduate students in Ontario 2024 – 2025

Toronto Metropolitan Graduate Fellowship .... 2025 – 2026
Awarded as a competitive, merit-based institutional fellowship to graduate students demonstrating academic excellence 2023 – 2024

Mathematics Graduate Award .... 2023 – 2026
Awarded by the department for outstanding academic performance in graduate mathematics

Professional Development Grant .... 2025 – 2026
Awarded by the department for academic dedication and research potential

Toronto Metropolitan Graduate Student Travel Fund Award 2025
Awarded by the institution to support research dissemination and active participation at academic conferences

Toronto Metropolitan Graduate Scholarship (declined) .... 2024 – 2025
Selected as a competitive institutional scholarship recipient through the Ontario Graduate Scholarship competition

Toronto Metropolitan Graduate Development Award .... 2024 – 2025
Awarded by the department for academic potential and professional success

York Graduate Fellowship .... 2021 – 2022
Awarded as an institutional fellowship supporting academic and research activities

Classical Studies Prize in Elementary Latin 2016
Awarded by the department to the student with the highest standing in Latin 1000

Class of '49 Prize 2010
Awarded by the department to the student with the highest mark in Calculus 1501

Western Scholarship of Excellence 2009
Recognized outstanding high school academic achievement as a merit-based entrance scholarship

Honours and Distinctions

Featured Research Article 2026/05

Mathematical Oncology Newsletter

- The article "How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection" was featured in *This Week in Mathematical Oncology*, Issue 370.

Presentations

A Gentle Introduction to Riemannian Gradient Descent Method on Smooth Manifolds

- *Chemistry Research Group Meeting (Invited Mini-lecture – Memorial University of Newfoundland)* 2026/01
- *Mathematical Biology Seminar (Mini-lecture – Toronto Metropolitan University)* 2025/10

How Mathematical Forms Describing Cytotoxic Action of Chemotherapy and Radiotherapy Bias Predictions

- *2025 Complex Systems Day (Talk – Toronto)* 2025/08
- *BIRS Workshop on Mechanistic Learning as a Combination of Machine Learning and Modelling in Mathematical Oncology (Poster – Banff)* 2025/01
- *Fields Institute Workshop on Mathematical Modelling of Cancer Treatments, Resistance, Optimization (Poster – Toronto)* 2024/09

Research Experience

Research Assistant 2019/07 – 2020/06

Krembil Research Institute Toronto, ON

- Topics:
  (1) Topological analysis on hydrogen bond networks of confined and interfacial waters
  (2) Implementation of a deep learning algorithm in indirect surface characterization
  (3) Effect of cholesterol on the structure of water at the surface of a model lipid membrane

Research Assistant 2015/05 – 2015/08

University of Western Ontario London, ON

- Topic: Molecular dynamics studies of nanosized aqueous droplets

Teaching Experience

Guest Instructor 2026/01 – 2026/05
Memorial University of Newfoundland St. John's, NL

- Designed and delivered intuition-focused lectures in general topology for a chemistry research group.
- Emphasized conceptual understanding rather than formal, proof-based understanding.

Teaching Assistant 2023/09 – Present
Toronto Metropolitan University Toronto, ON

- Led weekly labs guiding students through difficult problems and reviewing course material.
- Contributed to the development of exam marking schemes (MTH 231).

MTH 141 – Linear Algebra *MTH 207 – Calculus and Computational Methods I*
MTH 231 – Modern Mathematics II *MTH 240 – Calculus II*
MTH 330 – Calculus and Geometry *MTH 350 – Multivariate and Vector Calculus*
MTH 501 – Numerical Analysis I *MTH 430 – Dynamic Systems Differential Equations*
MTH 510 – Numerical Analysis *MTH 719 – Applied Linear Algebra*

Teaching Assistant 2021/09 – 2022/04
York University North York, ON

- Provided one-on-one and group tutoring to students

MATH 1013 – Applied Calculus I *MATH 1300 – Differential Calculus with Applications*
MATH 1025 – Applied Linear Algebra *MATH 1507 – Mathematics II*

Teaching Assistant 2013/09 – 2014/04
University of Western Ontario London, ON

- Led group conversation tutorials to support students in improving their Japanese speaking skills.
- Provided instructional support and supervision.

JAPANESE 1036 – Japanese for Beginners *JAPANESE 2260 – Intermediate Japanese*

Educational Outreach

Instructor 2026/02 – 2026/06
Toronto Garden Church North York, ON

- Taught mathematics to students in Grades 5–10 as part of a community outreach program.

Instructor 2025/03 – 2025/06

Toronto Milal Church Toronto, ON

- Taught essential computer skills to elderly learners as part of the senior computer literacy program, including the use of Microsoft Word, Paint, and introductory AI tools.

Instructor 2021/08 – 2024/07

Online Learning Community Korea

- Taught linear algebra, calculus, probability theory, and real analysis online to learners without strong mathematical backgrounds.
- YouTube Channel: *Math Friends*

Instructor 2015/09 – 2016/08

London Full Gospel Church London, ON

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

Instructor 2014/09 – 2015/08

Oakridge Presbyterian Church London, ON

- Taught Grade 9–12 mathematics to secondary school students as part of a community outreach program.

Instructor 2008/05 – 2009/07

Oakville Presbyterian Church Oakville, ON

- Taught mathematics and Japanese to immigrant secondary school students as part of a community outreach program.

Technical and Language Skills

- Programming Languages
    - Julia
    - MATLAB
    - Python
- Languages
    - Korean – Native
    - English – Advanced
    - Japanese – Advanced (JLPT N1)